EEE407 Tutorial 4c

1. A cubic water tank has surface area of 6.0 m^2 and is filled to 90% capacity six times daily. The water is heated from 20°C to 65°C. The losses per square metre of tank surface per 1°C temperature difference is 6.3 W. Find the loading in kW and the efficiency of the tank. Assume specific heat of water = 4200 J $kg^{-1}°C^{-1}$, water density = 997 kg/m^3, and one kWh = 3.6 MJ.

2. A 3-phase electric arc furnace has the following data:
   Current drawn = 5000 A; Arc voltage = 50 V
   Resistance of transformer referred to secondary = 0.002 Ω
   Reactance of transformer referred to secondary = 0.004 Ω
   a. Calculate the power factor and kW drawn from the supply.
   b. If the overall efficiency of the furnace is 65%, find the time required to melt 2 tonnes of steel if latent heat of steel = 37.25 kJ/kg, specific heat of steel = 444 J $kg^{-1}°C^{-1}$, melting point of steel = 1370°C and initial temperature of steel = 20°C.

3. Determine the efficiency of a high-frequency induction furnace which takes 10 minutes to melt 2 kg of an aluminum initially at a temperature of 20°C. The power drawn by the furnace is 5 kW, specific heat of aluminum = 887 J $kg^{-1}°C^{-1}$, melting point of aluminum = 660° C and latent heat of fusion of aluminum = 396 kJ/kg.

4. A low-frequency induction furnace has a secondary voltage of 20V and takes 600 kW at 0.6 p.f. when the hearth is full. If the secondary voltage is kept constant, determine the power absorbed and the p.f. when the hearth is half-full. Assume that the resistance of the secondary circuit is doubled but the reactance remains the same.

5. An insulating material 2 cm thick and 200 cm^2 in area is to be heated by dielectric heating. The material has relative permittivity of 5 and power factor of 0.05. Power required is 400 W and frequency of 40 MHz is to be used. Determine the necessary voltage and the current that will flow through the material. If the voltage were to be limited to 700 V, what will be the frequency to get the same loss?

6. A plywood board of 0.5 × 0.25 × 0.02 metre is to be heated from 25 to 125°C in 10 minutes by dielectric heating employing a frequency of 30 MHz. Determine the power required in this heating process. Assume specific heat of wood 1500 J $kg^{-1}°C^{-1}$; density of wood 600 kg/m^3 and efficiency of process 50%.